000660.KS - SK hynix

Full investment research | Model assumptions and source evidence

Framework 8.0 | Evidence and HBM execution risk reviewed October 8, 2026 | NAND pricing assumptions retained | Company research anchor: September 6, 2026

Status: HELD. Memory bottleneck with cyclical pricing.

Portfolio Base:88/100. Analytical score:53/60 =88/100.

Active NAND assumption: calendar Q3 2026 +23%; calendar Q4 2026 +20%; calendar Q1 2027 +15%. Company-level realization, allocation and persistence are modeled assumptions described below.

Valuation:6x forward-four-quarter operating profit. Active Q4 2027 price target:KRW 4,102,945.30 per share, pricing Q4 2027-Q3 2028.

Amounts: KRW trillions unless stated. All current forward tables represent one operating case. The 30% execution-risk probability enters the HBM operating inputs; no further probability multiplier is applied to total earnings or the valuation. Price-dependent ranking remains in Excel.

1. Thesis, products and competitive position

HELD. BOTTLENECK: qualified high-bandwidth memory integrates process technology, packaging, yield, customer co-development and reliable volume supply. Conventional DRAM and NAND add complementary inference demand. The listed entity is valued at the sanctioned 6x forward operating-profit multiple.

Why the product split matters

SK hynix combines HBM leadership and qualification with conventional DRAM and NAND manufacturing. HBM consumes advanced DRAM wafers and additional stacking, testing and packaging resources; its production economics cannot be inferred from conventional DRAM bit supply alone. The three-vector model isolates these businesses so that product mix changes flow into operating profit explicitly.

The economic bottleneck is qualified, yielding production delivered to customers. Process transitions, customer specifications, advanced packaging and capacity allocation determine whether strong demand converts to sales. Competitors' capacity announcements are not immediate interchangeable supply, but successful qualifications and rising yields can change the balance over time.

Contracts, capacity and shareholder attribution

The current evidence describes long-term agreements with customer-specific pricing and deposit provisions. It does not establish a uniform industry-wide floor-and-ceiling contract structure or a disclosed protected percentage for every product. Contract duration therefore supports planning without guaranteeing the modeled margin. Investment-related gains in net income are not recurring memory operating profit.

Indiana is a downstream packaging project for wafers produced in Korea. Its cleanroom and production milestones occur on different dates; the current schedule below places no incremental Indiana volume in the Q1 2028 price target window. Packaging an existing wafer is not a second chip sale.

The forecast share bridge incorporates issued shares and a conditional cancellation schedule. Announced repurchases, purchased treasury shares and completed cancellation are different events. The denominator sensitivity below makes failure to complete the assumed cancellation observable. The portfolio Base is 88,

consistent with the analytical category score.

Research provenance: the product and competitive analysis in this section synthesizes the supplied original research. Dated market-share, process and channel claims retain their original evidence status; this integration is not a new independent verification of those claims. Current earnings and model assumptions are set out below.

2. Earnings, guidance and operating assumptions

Latest earnings anchor

The Q2 company release reports preliminary consolidated revenue of KRW 79.3187T and operating profit of KRW 60.5426T. Net profit of KRW 93.9226T includes exceptional investment gains and is not the operating earnings input. The latest reported total is the calibration anchor; individual product allocations remain estimates. [SKH-1]

Management guidance and contract interpretation

Management guides Q3 DRAM bit growth around 10% and NAND low-single-digit growth; HBM4 ramps in H2 and HBM4E targets volume in 2027. Approximately ten LTAs typically span five years, with customer-specific pricing and deposits. Management does not disclose uniform price collars or contracted sales coverage. It attributes Q2 blended DRAM ASP effects to mix and shipments deferred into H2. [SKH-2]

The inference for the model is that greater demand visibility can support investment and resilient earnings, but does not mechanically justify a contractual floor under every product margin. SanDisk’s specific pricing collars are not imported.

Operating model and evidence limits

Three vectors carry separate revenue and operating margins: HBM, conventional DRAM, and NAND. OP equals the sum of the three revenue x margin terms. Gross-profit allocations are not published at this granularity, so the review does not present invented reported gross margins.

The active conditional model places HBM revenue at KRW38T in Q3 2026 and KRW66T in Q1 2028, with conventional DRAM at KRW 38T and KRW 51T, respectively. These forecasts are not management product guidance. Their exact unit/ASP and TAM-capture bridges remain unverified; the approved curve remains conditional and does not meet complete AH-12 evidentiary certification. The Q2 allocation reproduces the consolidated totals to rounding. Forecast confidence is LOW.

Margins peak in the modeled shortage period and then normalize as competing capacity qualifies. The curve follows the uploaded approved timing assumption; a factory announcement by itself neither proves oversupply nor warrants raising this path.

HBM4 acceptance, manufacturing integration and the active execution-risk branch

Damnang's October 7 research article describes concerns about SK hynix HBM4 assembly and reliability, including 8-high and 12-high stacks. The author links possible base-die, MR-MUF packaging, mechanical stress and high-speed signaling interactions. This is one attributed channel hypothesis, not a measured defect rate. Its strongest wording about all 12-high parts being defective is not adopted. The source's 20% unrecognized shipment assumption is a scenario input, not observed customer rejection. [SKH-10]

SK hynix's September 28 TSMC OIP exhibit showed Vera Rubin with 12-high 36GB HBM4, alongside HBM4E and 16-high HBM4 demonstrations. Its June 18 announcement described 12-high HBM4E samples at 16Gbps using Advanced MR-MUF. These are meaningful counter-evidence to universal technical failure.

Demonstrations and samples do not, however, establish stable good-unit yield, customer acceptance at scale or unit cost. Commercial execution risk remains a probabilistic model input. [SKH-11/12]

Key HBM4 specifications

Based on cited announcements

Category	Samsung	SK hynix	Micron
DRAM process	1c	1b	1β
Base die	In house 4nm	TSMC 12nm (reported)	In house CMOS
Speed per pin	11.7Gbps Up to 13Gbps	11.7Gbps 12 layer product announced	Above 11Gbps
Power efficiency gain	40%	40% or more	Above 20%

Power efficiency gains are versus each company's HBM3E; test conditions differ.
SK hynix power efficiency figures follow its official HBM4 announcement.
Speeds are published specifications and may differ from customer validation conditions for mass production.

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Figure SKH-10A. Announced HBM4 specifications reproduced by Damnang, October 7. Efficiency improvements use each vendor's own HBM3E baseline; the table is not a matched performance-per-watt or manufacturing-yield benchmark.

The engineering distinction is between signaling feasibility, stack-level reliability and saleable output. Reducing stack height may ease thermal and mechanical conditions, but it does not lengthen the bit interval at an unchanged interface speed. At 11Gbps the interval is about 91ps regardless of whether the stack is 8-high or 12-high. Base-die interface design, bumps, thermal paths, encapsulant and process conditions can interact; public product specifications do not identify which interface is failing. A roadmap or a wafer-start count is therefore not interchangeable with accepted good-stack capacity.

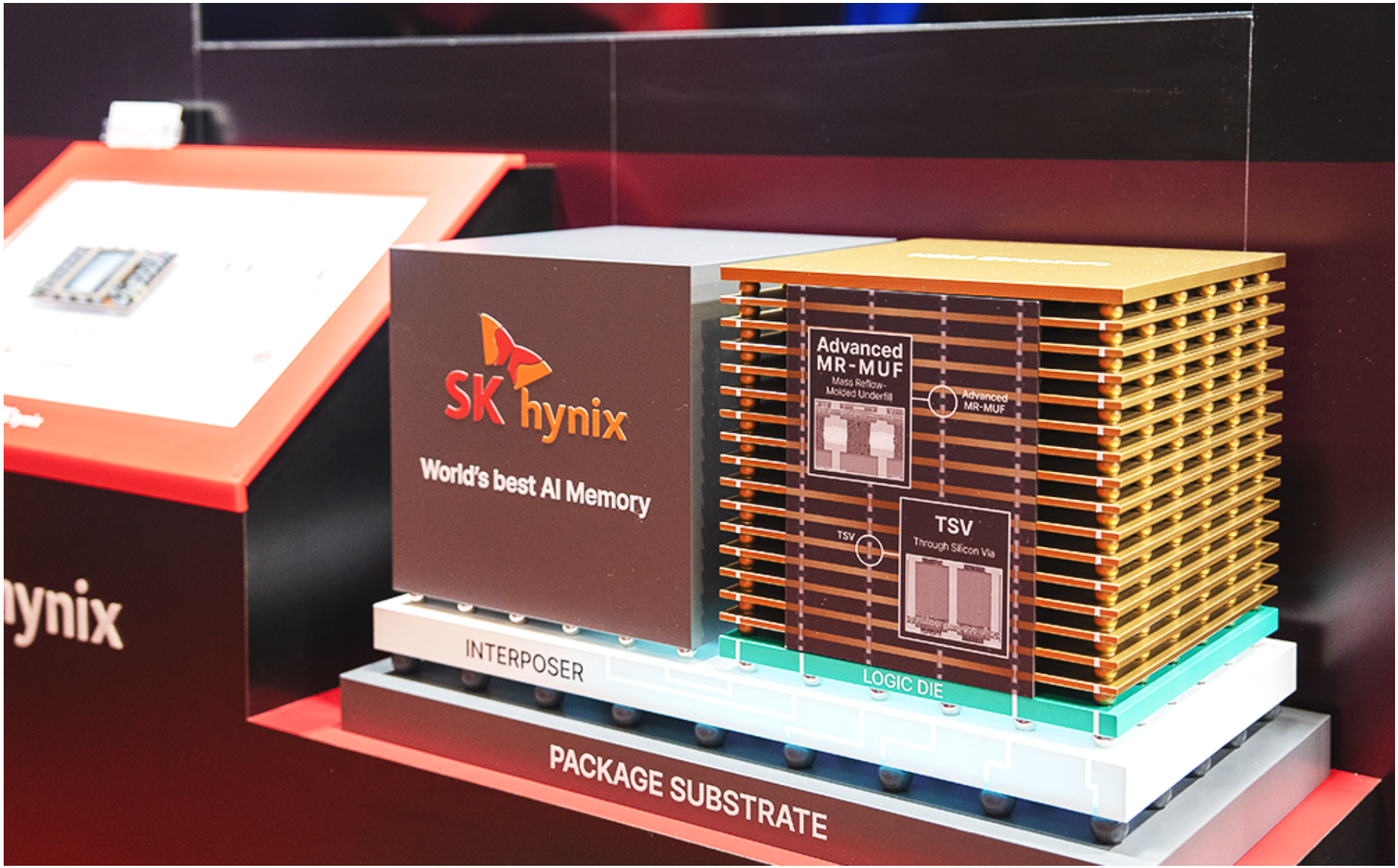


Figure SKH-10B. SK hynix HBM4 packaging concept reproduced in the source, originally from the 2025 TSMC symposium. It identifies the package elements; it does not measure an October 2026 defect mechanism or yield.

The active model assigns a **30% probability** to a temporary commercial execution shortfall affecting CY2027 HBM revenue. The chosen conditional endpoint is **7.5% of total HBM accepted revenue**, within a **5-10% bound**. This is deliberately below the author's 20% scenario because product demonstrations, potential remediation,

customer segmentation and HBM3E substitution limit the inference. The 7.5% is already net of substitution: another HBM3E credit or a separate wafer-yield loss would double-count the same event. It is an analyst proxy, not a reported HBM4 share or defect rate.

The bridge retains operating costs that do not disappear with missed acceptance. For reference HBM revenue R and operating margin m, conditional lost revenue L = R ×7.5%; avoidable cost = L ×(1-m) ×50%; extra rework and qualification expense = L ×2%. Probability-weighted HBM revenue is R-30%×L. HBM operating profit is R×m +30%×(-L + avoidable cost - extra expense). The model thus avoids only half the existing allocated cost attached to lost sales and retains the other half. This is a bounded cost proxy because actual product gross profit and fixed/variable expense splits are not disclosed.

The phase is active in all four CY2027 quarters. **Resolution from Q1 2028 is a model timing assumption; no catch-up sales are added.** CY2026 figures are not changed by a source scenario explicitly framed around 2027. Procurement reallocation or a prolonged qualification delay could extend the impact; the separate H1 2028 half-severity sensitivity below makes that risk visible without treating it as an independent event.

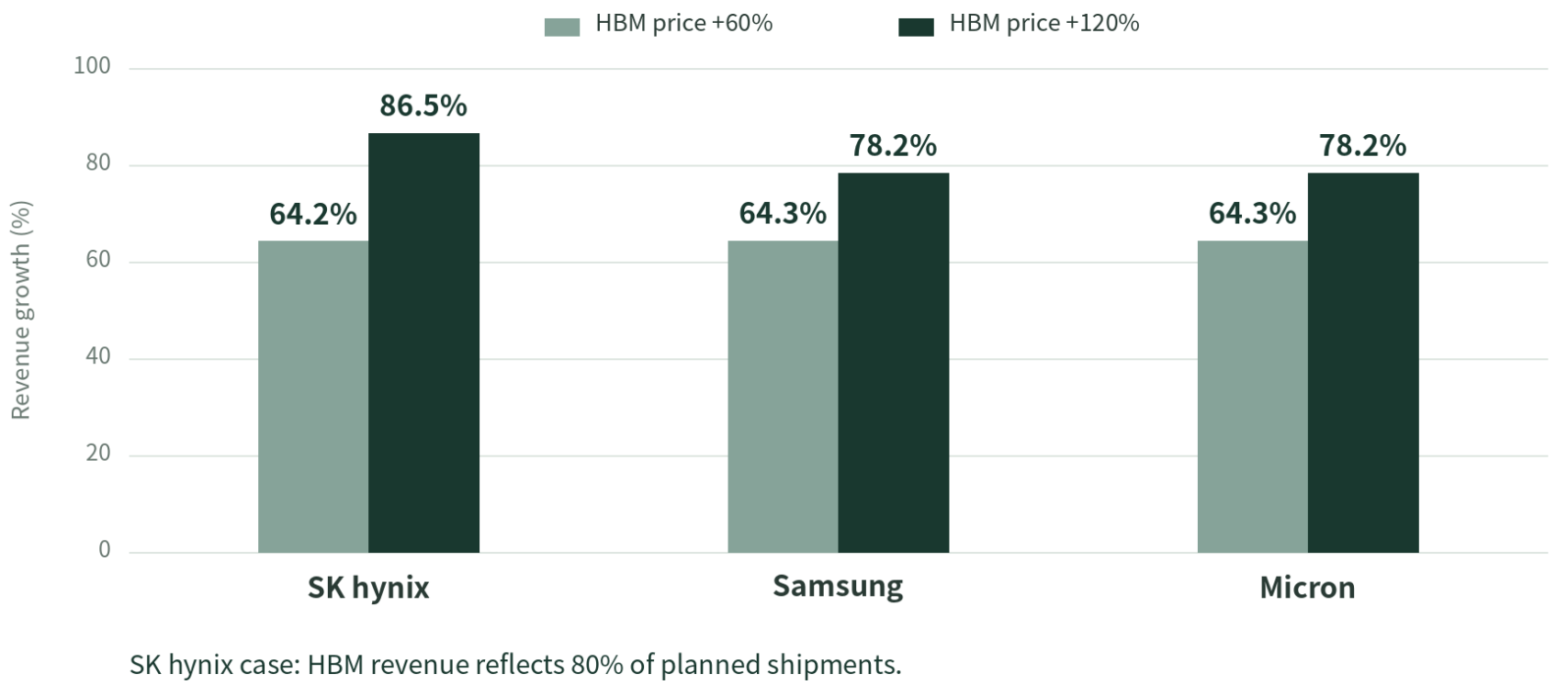
Quarter	Reference HBM sales	Conditional lost sales	Conditional cost avoided	Conditional rework	Active sales delta	Active OP delta
Q1 2027	50.000	3.750	0.304	0.075	-1.125000	-1.056375
Q2 2027	55.000	4.125	0.328	0.083	-1.237500	-1.163869
Q3 2027	59.000	4.425	0.350	0.088	-1.327500	-1.249178
Q4 2027	63.000	4.725	0.366	0.095	-1.417500	-1.335994

KRW trillions. The reference exposure is a bridge input, not a second active forecast. CY2027 active sales reduction is KRW5.107500T and OP reduction KRW4.805415T. Only Q4 2027 falls inside the active Q4 2027-Q3 2028 valuation window; the selected Q1 2028 resolution assumption explains the smaller horizon impact.

FIGURE 04 / MEMORY ECONOMICS

HBM shipment timing changes the benefit of higher prices

2027 DRAM revenue growth under two HBM price assumptions (%)



SK hynix case: HBM revenue reflects 80% of planned shipments.

Source: Damnang Research model.
Base revenue = 100 per company. HBM revenue mix: 30 / 15 / 15% (SK hynix / Samsung / Micron).
Commodity ASP +30%; commodity bits +15%; planned HBM bits +55%.
HBM shipment realization: SK hynix 80%; Samsung and Micron 100%.

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Figure SKH-10C. Damnang's normalized 2027 DRAM revenue scenario. Its 80% SK hynix shipment realization is an author assumption, not a verified yield. This table informs the risk direction; its company growth rates and 20% shortfall are not imported into the active model.

The source's Samsung and Micron growth comparison is not an independent company earnings forecast. It assumes HBM shares of 30% for SK hynix and 15% for Samsung/Micron, conventional bits +15%, conventional ASP +30%, HBM bits +55%, HBM ASP +60% or +120%, and SK hynix realizing 80% of planned HBM

shipments. Those assumptions produce roughly 64-86% versus 64-78% total DRAM revenue growth. They do not establish current customer allocations, gross margin or operating expense. The model uses one accepted-revenue risk branch instead of applying all those correlated assumptions again.

2028 DRAM revenue sensitivity

Change vs. 2027 under three commodity price scenarios

Commodity price	SK hynix	Samsung	Micron
Unchanged	+22.2%	+21.4%	+21.4%
Down 15%	+10.7%	+7.4%	+7.4%
Down 25%	+3.1%	−1.8%	−1.8%

Source: Damnang Research model. Rounded outputs.
Commodity bits +20%; HBM bits +40%; HBM ASP −10%.
2027 baseline uses HBM ASP +60% and the stated shipment realization assumptions.

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Figure SKH-10D. The source's 2028 DRAM revenue sensitivity, with conventional bits +20%, HBM bits +40% and HBM ASP -10%. Conventional ASP scenarios are flat/-15%/-25%; these are source-authored scenarios, not observed prices.

The conventional-memory profit pool can dominate revenue even when HBM execution attracts the most attention. That supports retaining separate conventional DRAM, HBM and NAND economics. The source's 2028 price sensitivity is not added on top of the model's existing normalization curve. Lower HBM customer concentration, reported shipment acceptance and substitution can matter more to cash earnings than peak data-rate leadership by itself.

Rule 16b ledger field	Current treatment
Economic probability	30%; one credible but unverified channel source, moderated by issuer demonstrations and uncertainty about affected products.
Conditional magnitude	7.5% net accepted-HBM revenue shortfall in CY2027; bounded5-10%. No measured yield claim.
Overlap adjustment	No separate acceptance factor existed in the reference product rows. HBM3E substitution is inside the chosen net shortfall. NAND pricing, generic score risk and additional customer transfers are excluded.

Rule 16b ledger field	Current treatment
Cost and recognition	Avoid 50% of allocated implied cost on lost revenue; retain balance; add 2% of conditional lost sales for rework. Revenue waits for acceptance.
Active timing	CY2027 only. ResolutionQ1 2028 assumed; no recovery revenue.
Unselected H1 2028 half-tail	Same risk continues at half severity for two more quarters: additional OP reduction 1.426815T; Q4 2027 ceiling KRW4,090,826.64 common / USD411.98 ADS. Not included in active tables.
Probability increases	Documented customer rejection, accepted-volume misses, repeated rework or remediation slipping into 2028; reassess exposure and timing together.
Probability decreases	Stable customer acceptance and volume-yield evidence across relevant SK hynix HBM4 programs; demonstrated remediation and economically effective substitution.
Protected valuation and score	Current Base 88 and analytical 53/60; parent 6x and ADS9x F4Q OP; existing share schedule, ADS ratio and FX. No global probability multiplier.

Share-count bridge and execution sensitivity

The August 19 filing states 730,492,365 issued common shares and an estimated 24,070,000 shares to cancel. Acquisition is planned for August 20-November 19, followed by cancellation in one batch; the actual quantity depends on execution. The model therefore uses 730.492365M in Q3 2026 and 706.422365M from Q4 2026, conditional on completion. [SKH-3]

The earlier F-1 reports March issued shares of 712.702365M, treasury shares of 4.405344M, and Q1 diluted weighted-average shares of 711.592641M. Its dilution note includes options and exchangeable bonds. Adding the IPO's 17.79M new common shares reconciles issued shares to the August filing. The review uses all issued shares as a fully distributed proxy, retaining existing treasury-share potential rather than deducting it again. It does not add historical weighted-average bond dilution on top of a fully distributed issued count without a current incremental issuance bridge. [SKH-6]

If the cancellation does not occur, the Q1 2028 price target is approximately 3.29% below the modeled result. Each 1M additional forward shares reduces it by roughly 0.14%. Additional grants or new issuance require a fresh denominator. This is a forward assumption, not a claim that cancellation is already complete.

Capital allocation and capacity timing

The company approved KRW 40T of repurchases and cancellation and plans to pursue returns exceeding 50% of cumulative 2025-2027 free cash flow. Further program details require later board decisions; those unapproved quantities are excluded. [SKH-4]

Indiana's cleanroom is scheduled for October 2028 and mass production for H2 2029. The facility packages wafers produced in Korea. It contributes no incremental revenue to the Q1 2028-Q4 2028 earnings window, and packaging the same wafer is not a second chip sale. The later HBM trajectory is not increased solely for this announcement. [SKH-5]

Affiliate construction, cash timing and attributable capacity

The October 6 DART amendment, approved by the board September 30, forecasts **KRW4.2462T of Q4 2026 purchases from SK ecoplant** for October 1-December 31. It comprises **KRW4.0049T of FAB construction** and **KRW0.2413T of supporting production facilities**. The original filing units are KRW millions. Its **4.89%** comparator uses **KRW86.852117T of FY2025 separate-company revenue**, not Q4 consolidated sales. [SKH-8]

The expected amount increased by at least 20% versus the February 27 disclosure. Crucially, the amendment includes new or changed work and **construction advances following changes in payment terms**, with a planned move from promissory notes toward cash. It is consequently material evidence about funding and project execution, while the portion representing additional physical construction remains unknown. It is neither all-new capital capacity nor cash already paid. The notice does not identify product capacity, site-level output or whether the work is incremental to the existing M15X/Yongin capital plan; it is not assigned to Indiana.

The current analyst judgment is **90% probability that the disclosed Q4 procurement broadly executes**, with the exact amount and timing still variable. An illustrative 90% × KRW4.2462T = **KRW3.82158T** measures execution exposure only; it is not an incremental capex, sales or OP forecast. On a comparable-scope interpretation of the at-least-20% increase, the previous estimate was at most **KRW3.5385T** and the revision at least **KRW0.7077T**. These are transaction-estimate bounds, not a physical-capex uplift, because advances and earlier cash payment can produce the change.

A broader **KRW0-4.2462T** overlap bracket tests the amount that could sit above the already modeled capital plan. It cannot be narrowed without the February 27 project baseline and a split among new scope, rephasing and advances. Even its upper endpoint cannot supply a defensible good-unit forecast: fab area and process allocation, equipment, commissioning, wafer starts, accepted yield, utilization and realized pricing are not disclosed. **This filing adds zero incremental modeled capacity, revenue and OP** because the capital proxy has no separately attributable output bridge and overlaps an existing growth plan to an unknown extent. The company-level operating path remains conditional; the zero is not a claim that construction has no economic value or that the source is unofficial.

Moving payments toward cash can increase near-term funding use without increasing the physical asset ultimately built. Advances are not automatically an immediate OP expense; later capitalization, commissioning and depreciation must be mapped before changing the operating-cost path. The same issuer exposure applies to 000660.KS and SKHY once. No second construction or HBM revenue pool is created by the ADS wrapper.

SK hynix | Q4 affiliate purchases

October 6, 2026 DART amendment | expected October–December 2026 | KRW trillions

DISCLOSED EXPECTED PURCHASE COMPONENTS

FAB construction 4.0049

Supporting facilities 0.2413 | Total 4.2462

ILLUSTRATIVE EXECUTION EXPOSURE

90% × expected purchases = 3.82158

The 90% judgment concerns broad Q4 procurement execution. The weighted amount is an illustration, not a forecast of new cash capex, productive capacity or revenue.

INCREMENT ABOVE EXISTING CAPITAL PLAN: 0–4.2462

The revision includes construction advances and payment-term changes toward cash. Physical scope and capital-plan overlap are unknown. No attributable wafer/bit-output bridge supports additional modeled OP.

Source: DART rcpNo 20261006000253, board decision 2026-09-30 | SKH-8

SK hynix Q4 affiliate-purchase mix and illustrative execution exposure

Figure SKH-8. Source: October 6 DART amendment. The component amounts are company expectations, while the 90% probability and overlap bracket are analyst judgments. Payment advances and budget overlap prevent treating this as additional physical capacity.

Figure input or calculation	KRW trillions, except probability	Basis
FAB construction	4.0049	Expected Q4 purchases
Supporting production facilities	0.2413	Expected Q4 purchases
Total	4.2462	Sum of disclosed components
Broad Q4 execution judgment	90%	Analyst probability
Illustrative execution exposure	3.82158	4.2462 ×90%; no active cash-budget input
Potential amount above existing plan	0-4.2462	Unresolved overlap bracket

Early memory architecture evidence

An anonymous SK hynix engineer's September 19 interview, received as a Damnang newsletter preview on October 6, discusses cell/peripheral separation, defect inspection and process coordination for future DRAM. This is an author-edited partial preview, not a retrieved full article or an issuer production roadmap. Wafer bonding in that context does not confirm an HBM stack height or validate the earlier 12-high rumor. [SKH-9]

The subjective probability of a **measurable commercial benefit to SK hynix during the active Q4 2027-Q3 2028 valuation window is 10%**, reflecting early architecture/process progress without a production date. The attempted earnings bridge is exposed wafer starts × change in saleable bits per wafer × ASP, less bonding, inspection and yield-related costs. Neither the exposed node/wafer count nor density, yield or cost deltas is identified, so the current preview cannot support defensible bounded company dollars. Its active revenue/OP increment is zero for that specific unquantifiability reason, not because official confirmation is required. Full text, original images and production-node/timing evidence remain requested from DOT. The owner-deferred UBS snapshot remains unchanged and is not activated by this separate newsletter.

Active NAND pricing and transmission

The active NAND benchmark is **calendar Q3 2026 +23%, Q4 2026 +20% and Q1 2027 +15% sequentially**. Q3/Q4 retain the upper bounds of the owner-supplied TrendForce ranges, ranked Tier 1 alongside Serenity. The added Q1 forecast comes from the owner-supplied Jeff Pu/Jukan summary citing TrendForce. The owner authorized this combined case on September 21. The quoted analyst path was +21%/+17%/+15%; the active case retains the owner's higher 2026 selections. [TF-1, JF-1]

With Q2 2026 indexed to 1.000, active calendar price indexes are **1.230 in Q3, 1.476 in Q4 and 1.6974 in Q1 2027**. The Q1 level is 69.74% above Q2 2026. The +15% is applied once in Q1; it does not recur every quarter. Later forecasts retain their existing operating trajectory at this higher benchmark level. Persistence through 2030 is a modeling assumption, not a forecast supplied by TrendForce or the quoted analysts.

The inherited company curves did not contain an auditable NAND ASP/bit split. The current calculation therefore uses **reference-price calibration divisors of 1.150 for Q3 and 1.2075 for Q4 onward**. These reproduce the supplied curve at the previous upper-bound benchmark and are calculation inputs, not a second active price forecast. Reference-price NAND revenue = calibration NAND revenue / reference index. Active NAND revenue = reference-price NAND revenue x active index. The

underlying volume/mix proxy is held constant during this repricing; it is not a measured bit shipment series.

This full-transmission operating case replaces the price component rather than applying another 23% and 20% on top of already priced revenue. There is no corresponding increase to DRAM or HBM pricing, no new fab capacity, and no automatic extra buyback. Company contracts, qualification, mix and timing remain the tests of realized earnings.

SK hynix and Solidigm NAND economics

NAND includes the existing SK hynix/Solidigm exposure; Solidigm is not added as another revenue vector. The HBM and conventional DRAM paths retain their individual margins. NAND operating cost = calibration NAND revenue x (1 - calibration NAND OPM). That total cost, including the embedded production and operating expense assumptions, is retained in absolute KRW while NAND revenue is repriced. Active NAND OP equals active NAND revenue minus this cost, and active NAND OPM is derived.

The resulting incremental price revenue flows one-for-one to operating profit in the price target case. Extra profit-linked compensation, cost inflation, customer concessions, volume loss or adverse mix would reduce that conversion. This is an explicit price sensitivity assumption, not a company commitment to unchanged cost. The current plus next three quarters of issuer OP and the same issuer share schedule support both listings. SKHY's separate valuation multiple does not create extra issuer earnings.

3. Analytical score

The category arithmetic gives **53/60 = 88/100** after the existing -3-point risk deduction. The portfolio Base is **88/100**. Risk-class detail appears in section 7.

Category	Points	Max.	Basis
Core thesis	19	20	HBM co-development, yield and packaging integration create substantial switching barriers.
Cycle stage	10	10	Demand exceeds qualified supply; HBM4 production conversion underway.
TAM expansion	10	10	HBM, server DRAM and workload-specific storage all expand with inference.
Catalysts	5	5	HBM4/HBM4E, M15X/Yongin and approved cancellation.
Macro	10	10	AI memory intensity and capacity lead times support demand visibility.
Alpha	2	5	Contract and share-filing detail improves understanding, but the core thesis is widely covered.
Risks	-3	0	Competitive supply/pricing; product and capacity execution; customer concentration and contract enforceability.

4. Product economics and quarterly operating model

All forecast rows below are **Model**, with LOW confidence in company realization. Amounts are KRW trillions. The historical consolidated anchor and management guidance remain distinct in section 2. Product OP is used because gross-profit and opex allocations are not reported at this product granularity.

Vector evidence

Vector	Source / derivation	Confidence
HBM	Product exposure anchored to the supplied quarterly assumptions;30% CY2027 acceptance-risk branch with net 7.5% conditional shortfall and explicit cost bridge. No NAND repricing.	LOW forecast
Conventional DRAM	Supplied quarterly operating assumptions; earnings anchor in section 2. No NAND repricing applied to this vector.	LOW forecast
NAND including Solidigm	Supplied quarterly operating assumptions; earnings anchor in section 2. Explicit TF-1 NAND price bridge below.	LOW forecast

NAND price and revenue derivation

The reference-price revenue column is the active model's volume/mix proxy in Q2-price units, not another forecast. The reference divisor is a calibration input only.

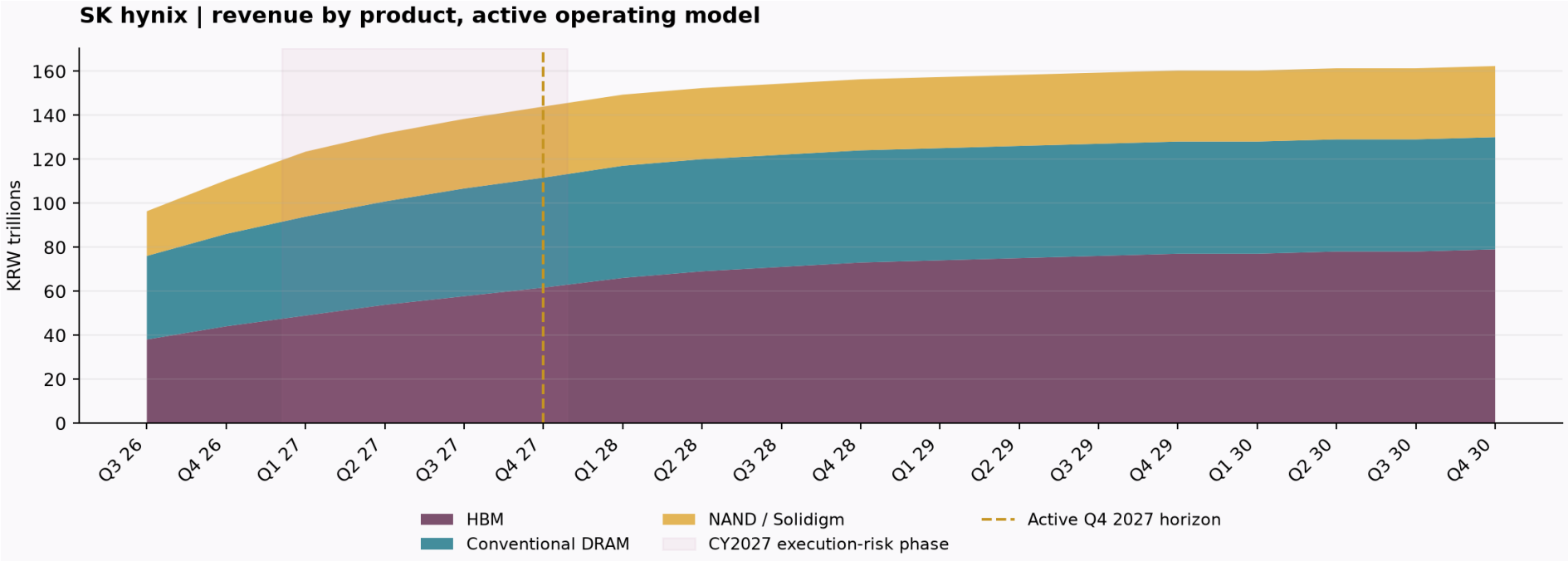
Quarter	Reference divisor	Active price index	Reference-price NAND revenue	Active NAND revenue
Q3 2026	1.1500	1.2300	16.52	20.32
Q4 2026	1.2075	1.4760	16.56	24.45
Q1 2027	1.2075	1.6974	17.39	29.52
Q2 2027	1.2075	1.6974	18.22	30.93
Q3 2027	1.2075	1.6974	18.63	31.63
Q4 2027	1.2075	1.6974	19.05	32.33
Q1 2028	1.2075	1.6974	19.05	32.33
Q2 2028	1.2075	1.6974	19.05	32.33
Q3 2028	1.2075	1.6974	19.05	32.33
Q4 2028	1.2075	1.6974	19.05	32.33
Q1 2029	1.2075	1.6974	19.05	32.33
Q2 2029	1.2075	1.6974	19.05	32.33
Q3 2029	1.2075	1.6974	19.05	32.33
Q4 2029	1.2075	1.6974	19.05	32.33

Quarter	Reference divisor	Active price index	Reference-price NAND revenue	Active NAND revenue
Q1 2030	1.2075	1.6974	19.05	32.33
Q2 2030	1.2075	1.6974	19.05	32.33
Q3 2030	1.2075	1.6974	19.05	32.33
Q4 2030	1.2075	1.6974	19.05	32.33

NAND operating cost and profit

Quarter	NAND revenue	NAND operating cost	NAND OP	NAND OP margin
Q3 2026	20.32	6.23	14.09	69.3%
Q4 2026	24.45	6.32	18.13	74.1%
Q1 2027	29.52	6.43	23.09	78.2%
Q2 2027	30.93	6.60	24.33	78.7%
Q3 2027	31.63	6.64	24.99	79.0%
Q4 2027	32.33	6.67	25.66	79.4%
Q1 2028	32.33	6.90	25.43	78.7%
Q2 2028	32.33	7.24	25.09	77.6%
Q3 2028	32.33	7.70	24.63	76.2%
Q4 2028	32.33	8.28	24.05	74.4%
Q1 2029	32.33	8.92	23.41	72.4%
Q2 2029	32.33	9.52	22.81	70.5%
Q3 2029	32.33	10.07	22.26	68.8%
Q4 2029	32.33	10.58	21.75	67.3%
Q1 2030	32.33	10.99	21.34	66.0%
Q2 2030	32.33	11.34	20.99	64.9%
Q3 2030	32.33	11.64	20.69	64.0%
Q4 2030	32.33	11.87	20.46	63.3%

Product revenue build

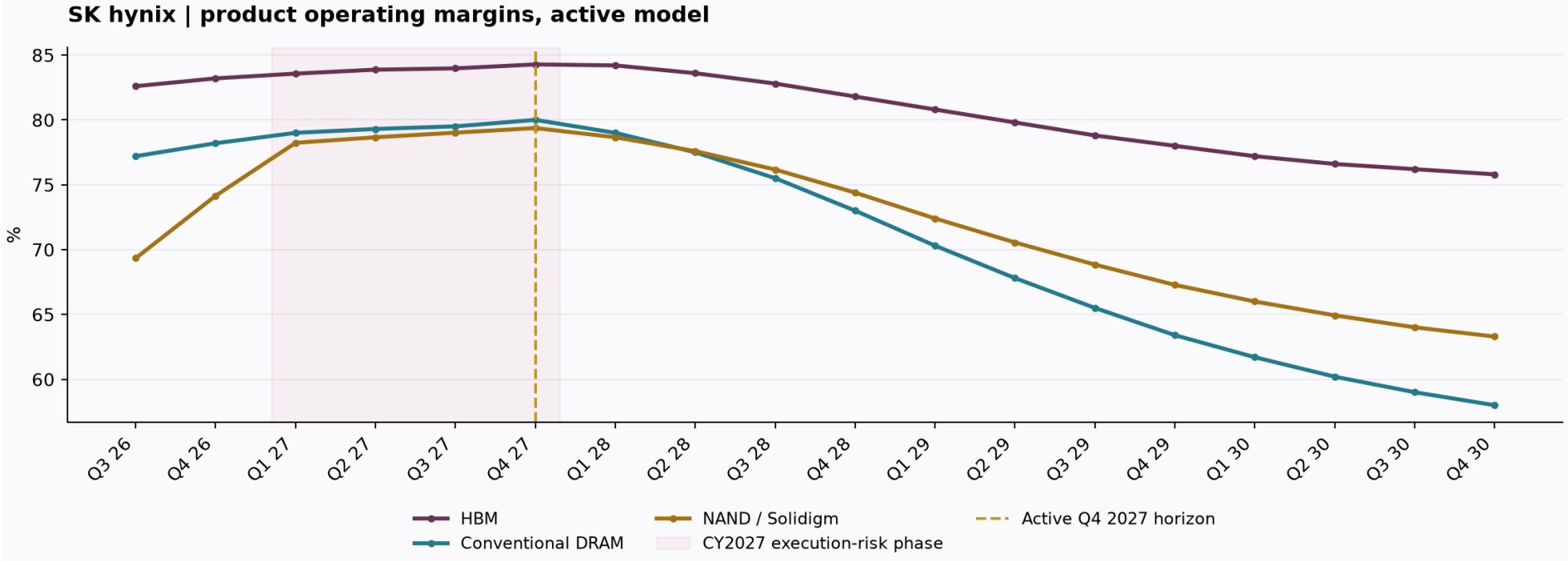


Active forecast revenue by product. CY2027 includes the probability-weighted accepted-HBM shortfall. Table below retains complete numerical values.

Quarter	HBM	Conventional DRAM	NAND including Solidigm	Total
Q3 2026	38.00	38.00	20.32	96.32
Q4 2026	44.00	42.00	24.45	110.45
Q1 2027	48.88	45.00	29.52	123.39
Q2 2027	53.76	47.00	30.93	131.69
Q3 2027	57.67	49.00	31.63	138.30
Q4 2027	61.58	50.00	32.33	143.91
Q1 2028	66.00	51.00	32.33	149.33
Q2 2028	69.00	51.00	32.33	152.33
Q3 2028	71.00	51.00	32.33	154.33

Quarter	HBM	Conventional DRAM	NAND including Solidigm	Total
Q4 2028	73.00	51.00	32.33	156.33
Q1 2029	74.00	51.00	32.33	157.33
Q2 2029	75.00	51.00	32.33	158.33
Q3 2029	76.00	51.00	32.33	159.33
Q4 2029	77.00	51.00	32.33	160.33
Q1 2030	77.00	51.00	32.33	160.33
Q2 2030	78.00	51.00	32.33	161.33
Q3 2030	78.00	51.00	32.33	161.33
Q4 2030	79.00	51.00	32.33	162.33

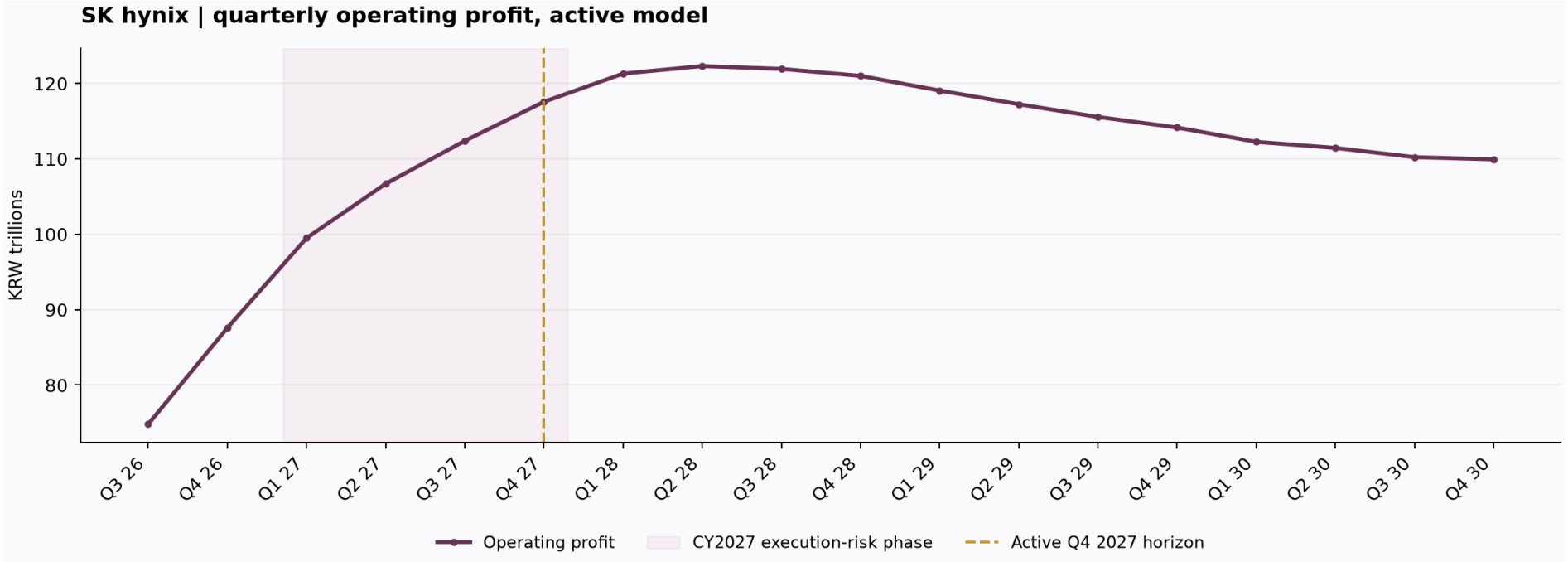
Product operating margins



Active forecast product operating margins. HBM retains fixed burden and rework expense in the CY2027 risk phase.

Quarter	HBM	Conventional DRAM	NAND including Solidigm
Q3 2026	82.6%	77.2%	69.3%
Q4 2026	83.2%	78.2%	74.1%
Q1 2027	83.6%	79.0%	78.2%
Q2 2027	83.9%	79.3%	78.7%
Q3 2027	84.0%	79.5%	79.0%
Q4 2027	84.3%	80.0%	79.4%
Q1 2028	84.2%	79.0%	78.7%
Q2 2028	83.6%	77.5%	77.6%
Q3 2028	82.8%	75.5%	76.2%
Q4 2028	81.8%	73.0%	74.4%
Q1 2029	80.8%	70.3%	72.4%
Q2 2029	79.8%	67.8%	70.5%
Q3 2029	78.8%	65.5%	68.8%
Q4 2029	78.0%	63.4%	67.3%
Q1 2030	77.2%	61.7%	66.0%
Q2 2030	76.6%	60.2%	64.9%
Q3 2030	76.2%	59.0%	64.0%
Q4 2030	75.8%	58.0%	63.3%

Quarterly operating model



Active forecast quarterly operating profit. One underlying issuer model supports both listings.

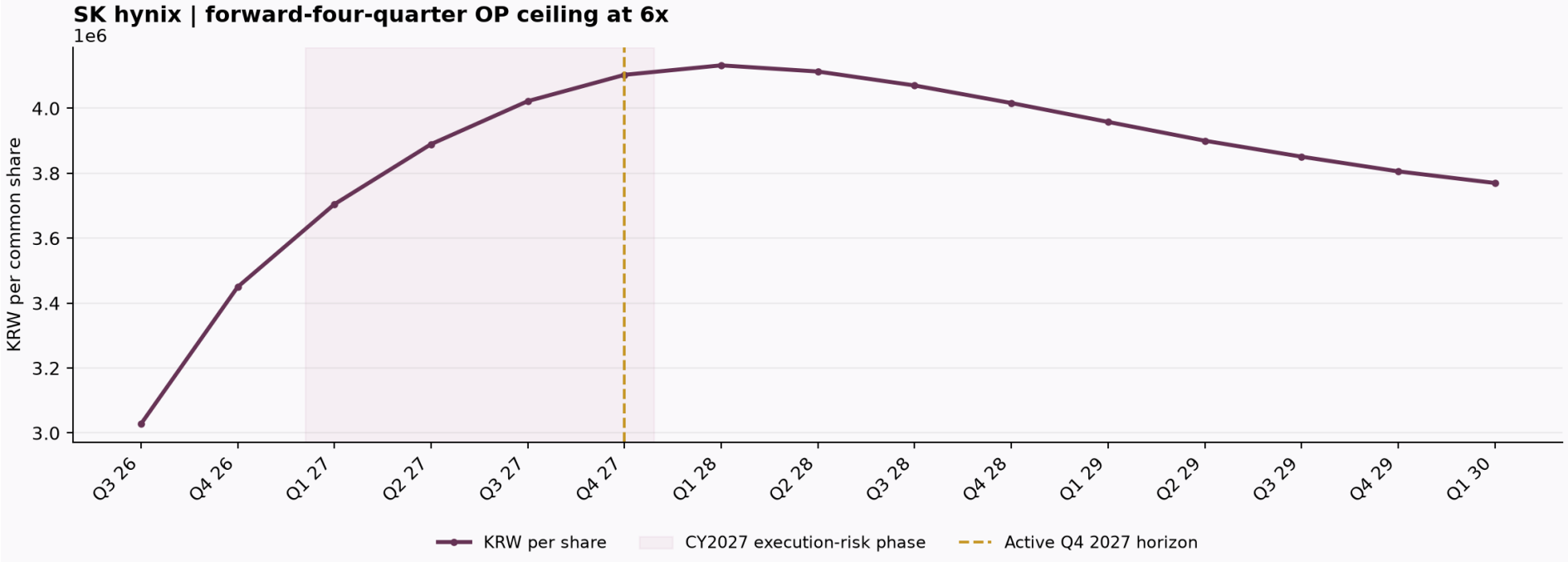
Quarter	Status	Revenue	Operating profit	OP margin
Q3 2026	Model	96.32	74.81	77.7%
Q4 2026	Model	110.45	87.58	79.3%
Q1 2027	Model	123.39	99.49	80.6%
Q2 2027	Model	131.69	106.69	81.0%
Q3 2027	Model	138.30	112.37	81.3%
Q4 2027	Model	143.91	117.56	81.7%
Q1 2028	Model	149.33	121.29	81.2%
Q2 2028	Model	152.33	122.30	80.3%
Q3 2028	Model	154.33	121.92	79.0%

Quarter	Status	Revenue	Operating profit	OP margin
Q4 2028	Model	156.33	121.00	77.4%
Q1 2029	Model	157.33	119.05	75.7%
Q2 2029	Model	158.33	117.24	74.0%
Q3 2029	Model	159.33	115.55	72.5%
Q4 2029	Model	160.33	114.15	71.2%
Q1 2030	Model	160.33	112.25	70.0%
Q2 2030	Model	161.33	111.44	69.1%
Q3 2030	Model	161.33	110.22	68.3%
Q4 2030	Model	162.33	109.93	67.7%

Annual model cross-check

Model year	Revenue	Operating profit	OP margin
2027	537.30	436.11	81.2%
2028	612.33	486.50	79.5%
2029	635.33	465.99	73.3%
2030	645.33	443.84	68.8%

5. Forward four-quarter operating price target



000660.KS active ceiling path,6x F4Q OP; conditional share schedule retained.

Price target (KRW/share) = forward four-quarter OP (KRW trillions) x 6 / forward common shares (millions) x 1,000,000. Shares are 730.492365M for Q3 2026 and 706.422365M from Q4, conditional on the announced cancellation. The model does not treat cancellation as already completed.

Every row prices its stated quarter plus the next three quarters. No additional portfolio-wide probability haircut, tax conversion, net-debt addition or EPS multiple is inserted; the execution-risk probability is already inside HBM operating inputs. These are the portfolio's sanctioned operating-profit valuation conventions. The final three operating quarters support the preceding windows and have no standalone price target because later earnings are unavailable.

Price target quarter	Exact earnings window	Forward-4Q OP	Multiple	Economic shares M	Price target KRW/share
Q3 2026	Q3 2026 to Q2 2027	368.57	6.00	730.49	3,027,287.61
Q4 2026	Q4 2026 to Q3 2027	406.13	6.00	706.42	3,449,462.44
Q1 2027	Q1 2027 to Q4 2027	436.11	6.00	706.42	3,704,108.09
Q2 2027	Q2 2027 to Q1 2028	457.92	6.00	706.42	3,889,315.73
Q3 2027	Q3 2027 to Q2 2028	473.52	6.00	706.42	4,021,878.77

Price target quarter	Exact earnings window	Forward-4Q OP	Multiple	Economic shares M	Price target KRW/share
Q4 2027	Q4 2027 to Q3 2028	483.07	6.00	706.42	4,102,945.30
Q1 2028	Q1 2028 to Q4 2028	486.50	6.00	706.42	4,132,120.43
Q2 2028	Q2 2028 to Q1 2029	484.26	6.00	706.42	4,113,086.49
Q3 2028	Q3 2028 to Q2 2029	479.20	6.00	706.42	4,070,126.36
Q4 2028	Q4 2028 to Q3 2029	472.84	6.00	706.42	4,016,031.24
Q1 2029	Q1 2029 to Q4 2029	465.99	6.00	706.42	3,957,850.75
Q2 2029	Q2 2029 to Q1 2030	459.18	6.00	706.42	3,900,060.96
Q3 2029	Q3 2029 to Q2 2030	453.39	6.00	706.42	3,850,841.11
Q4 2029	Q4 2029 to Q3 2030	448.06	6.00	706.42	3,805,562.25
Q1 2030	Q1 2030 to Q4 2030	443.84	6.00	706.42	3,769,719.67

Derivation example: active Q4 2027

117.560435 + 121.293429 + 122.295429 + 121.919429 = 483.068721T forward-four-quarter OP. Apply the 6x multiple and 706.422365M economic shares to obtain **KRW 4,102,945.30 per share**. All calculations retain full precision; displayed table totals can differ from rounded components.

Q1 2028 mechanical share diagnostic

This retained diagnostic holds the Q1 2028 operating window fixed. It is separate from the active Q4 2027 price target and does not revise the share forecast.

Q1 2028 denominator	Price target KRW/share
Current model	4,132,120.43
5% more shares	3,935,352.79
10% more shares	3,756,473.12

6. Industry relationships and read-throughs

TAM expansion and cross-book learning

HBM supports accelerator compute; server DRAM supports agent working memory; SSDs support data lakes and cache tiers. These are distinct product demand pools, but an entire data-center investment budget cannot be counted as chip revenue. Follow workload mix and memory content, not just GPU counts.

Apply the contract lesson to Samsung, Micron and SanDisk: agreement term, minimum volumes, deposits, price floors and price ceilings must be checked separately for each issuer. Apply the Indiana timing lesson to equipment and packaging suppliers: distinguish equipment installation, cleanroom readiness, qualification and revenue production.

Use the same issuer-level operating and share assumptions for 000660.KS and SKHY. Compare Samsung, Micron, Kioxia and SanDisk by product and specific contract disclosure; do not import one company's pricing terms. HBM demand also informs packaging and test suppliers, subject to supplier-specific awards and timing.

The active NAND benchmark is shared with the SNDK and Kioxia updates, but issuer exposure, recognition dates and operating cost treatment differ. The same percentage increase in NAND prices does not imply the same percentage increase in consolidated earnings or per-share value.

7. Risks and falsification tests

The following ledger makes the existing aggregate risk deduction explicit. It does not introduce another deduction or change the analytical score.

Risk class	Existing deduction
Competitive supply, pricing and product share	-1
Process, packaging and capacity execution	-1
Customer concentration and contract enforceability	-1
Total	**-3**

Risks, score and triggers

Risks are organized into three classes: competitive capacity/pricing, production execution, and customer/contract dependence. The score is thesis quality, while the forward price target remains conditional operating math.

Next checks: accepted HBM4 volume, packaging rework, customer qualification and remediation timing; the Q3 earnings release, repurchase completion and cancellation filing, HBM4 yield/volume conversion, HBM4E qualification, and an explicit product units/ASP bridge. Reconcile the October 6 affiliate-purchase amendment to the February 27 baseline, separate new construction from advances, and check cash payments, commissioning and accepted output before assigning any additional earnings. A full product-level rebuild remains necessary before describing the long-range operating path as fully validated.

Numbered monitoring tests

#	Checkpoint	Evidence supporting the operating case	Counter-test / failure signal
1	Product economics	Reconcile HBM, DRAM and NAND mix to consolidated OP.	Investment gains or unexplained product residuals do not validate recurring profitability.
2	Agreement terms	Track disclosed coverage, deposits, pricing and customer performance.	Customer renegotiation or non-uniform terms defeats a universal collar assumption.
3	Capacity	Check qualified output and the separate Indiana milestones.	A cleanroom milestone alone does not establish packaged output or another chip sale.
4	Shares	Reconcile purchased, treasury, issued and cancelled shares.	Incomplete cancellation requires the denominator sensitivity rather than treating announced retirement as complete.

NAND pricing tests

Checkpoint	Required observation	Failure signal
Realized ASP	Track calendar Q3 +23%, Q4 +20% and Q1 2027 +15% benchmark against company NAND mix and realized ASP.	Contract caps, timing or discounts cause weaker transmission.
Output and cost	Reconcile saleable bits, mix, costs and incentive expense to the fixed reference-price curve.	Price gains are absorbed by lower units or additional costs.
Persistence	Test whether Q1 price gains persist as assumed in later forecasts.	ASP cuts or new capacity erase the retained price premium.

8. Sources, evidence gaps and model snapshot

Sources below are those documented in the supplied earnings review. LOW-confidence product allocations, future margins, capacity bridges and incomplete management-call records remain unresolved where stated. The current review follows Framework 8.0. The inherited product unit/ASP bridges and stated source gaps remain unresolved; an evidence update does not establish complete evidentiary certification.

- **SKH-1** · 2026-07-29 · Company Q2 financial results. Consolidated preliminary earnings anchor.
- **SKH-2** · 2026-07-29 company event · Q2 management transcript. Full management presentation and Q&A, including non-uniform LTA pricing.

- **SKH-3** · 2026-08-19 · Company cancellation Form 6-K. Issued shares, cancellation estimate and acquisition window.
- **SKH-4** · 2026-08-19 · Company shareholder-return announcement. Approved cancellation and distinction from prospective additional returns.
- **SKH-5** · 2026-08-28 · Company Indiana groundbreaking announcement. Cleanroom October 2028, mass production H2 2029.
- **SKH-6** · 2026-07-06 · Company amended F-1, financial notes 19 and 27. Pre-IPO shares, treasury stock, options and exchangeable-bond dilution; local full filing read.
- **SKH-7** · 2026-09-01 · Uploaded model and artifact - User upload: v3_2_master_portfolio(1).xlsx; 000660KS_SKHynix_DeepDive.md. Sanctioned 6x multiple and inherited conditional product path.

Original source files are retained in the external version archive.

- **SKH-8** · 2026-10-06 · DART amendment: large affiliate transactions. September 30 board decision covering October-December 2026 expected purchases from SK ecoplant. KRW4.2462T includes construction advances and payment-term changes; prior comparable amount/project mapping is unresolved. Original Korean HTML/text retained and hash-verified.
- **SKH-9** · Interview September 19; preview received October 6, 2026 · Damnang, The Next Memory Race Will Not Be Decided by Scaling Alone. Anonymous engineer, personal views and author-edited partial preview; exact publication time, full article and original body images were not retrieved. Qualitative architecture evidence with a 10% subjective in-window commercial-benefit judgment; no measurable company-dollar bridge.
- **TF-1** - Owner-supplied TrendForce revision, incorporated September 20, 2026: Q3 NAND ASP +18-23% and Q4 +15-20% sequentially. Active selections: +23% and +20%. The exact revised publication/table was not independently retrieved. Source priority is Tier 1 alongside Serenity as instructed by the owner.
- **TF-2** - August 18, 2026, TrendForce NAND supplier research. Samsung NAND revenue estimate $23.06B; SK hynix Group includes Solidigm. The article's aggregate and individual supplier totals do not reconcile; its aggregate and market-share denominator are not used in the model.
- **TF-3** - September 7, 2026, TrendForce DRAM supplier research. Samsung Q2 DRAM revenue estimate $60.98B, including the DRAM product mix. The Samsung allocation uses matched Q2 product estimates in USD, avoiding a new FX conversion.
- **JF-1** - Supplied September 21, 2026: Jeff Pu (@sssjeffpu) and Jukan (@jukan05), "NAND Pricing Breakout Driven by eSSD" summary in the owner's conversation. Quoted Q3 2026 / Q4 2026 / Q1 2027 blended NAND changes: +21% / +17% / +15%, citing TrendForce demand research. Only the Q1 +15% is newly adopted; the owner retains +23% and +20% for 2026. This is an owner-authorized combined case. The original analyst note and exact TrendForce table were not independently retrieved. Enterprise demand commentary is contextual and adds no shipment volume to this update.
- **SKH-10** · October7,2026 · Damnang, How HBM4 Is Changing Competition Among the Three Memory Companies. Full local newsletter record219245879; DOTNEWS-20261007-1130 revision2. Source scenarios and claims are attributed, with all8 research images inspected. Author scenarios are not issuer guidance.
- **SKH-11** · September28,2026 · SK hynix TSMC OIP exhibition. Vera Rubin display with12-high36GB HBM4; product demonstrations counter blanket failure but do not disclose accepted-volume yield.
- **SKH-12** · June18,2026 · SK hynix HBM4E sample announcement.16Gbps12-high samples using Advanced MR-MUF; samples are distinct from mass production economics.

Model snapshot: Framework8.0, October8,2026. One active operating model with 30% CY2027 HBM acceptance-risk probability and retained NAND pricing. Portfolio Base 88/100; analytical 53/60.18 operating quarters and15 complete forward-four-quarter windows. Active Q4 2027 price target:KRW 4,102,945.30 per share. Current horizon and all valuation/share conventions retained.